



VENICE SUMMER SCHOOL IN DIGITAL & PUBLIC HUMANITIES · LECTURE

Perception and biases from La Biennale di Venezia's International Art Exhibition

Reading the artworks working with generative algorithms

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From mediated narratives to the AI-Museum

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03 The Nexus — abduction & agency in the AI-Museum

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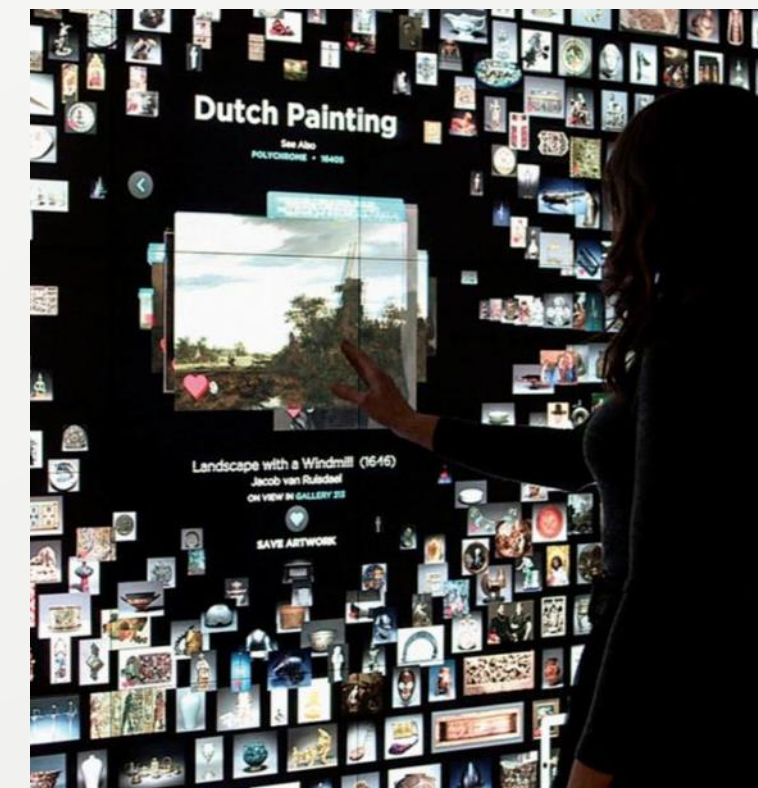
The AI turn in cultural practice

Experiments in which technology becomes a tool for reflecting on *identity* and *perception* anticipate a broader transformation in the relationship between artificial intelligence and cultural practice.

It is within this horizon of human–machine hybridisation that the museum world has begun to recognise digital technologies as means of **knowledge** and **relation**.

KEY MOVE

Digitisation campaigns produce a mass of data machines can process — offering research material to the Digital Humanities and receiving renewed impulse from them.



TOWARD THE POST-MUSEUM

A *post-museum* emerges — a process or experience that takes on different architectural forms and centres on the spaces, concerns and ambitions of communities. The theme was reaffirmed at NEMO's conference *Innovation and Integrity: Museums Paving the Way in an AI-driven Society* (20 March 2024).

"Museums are excellent places to negotiate technological developments together with the public, and offer spaces to learn, experiment and build knowledge."

— NEMO, Network of European Museum Organisations

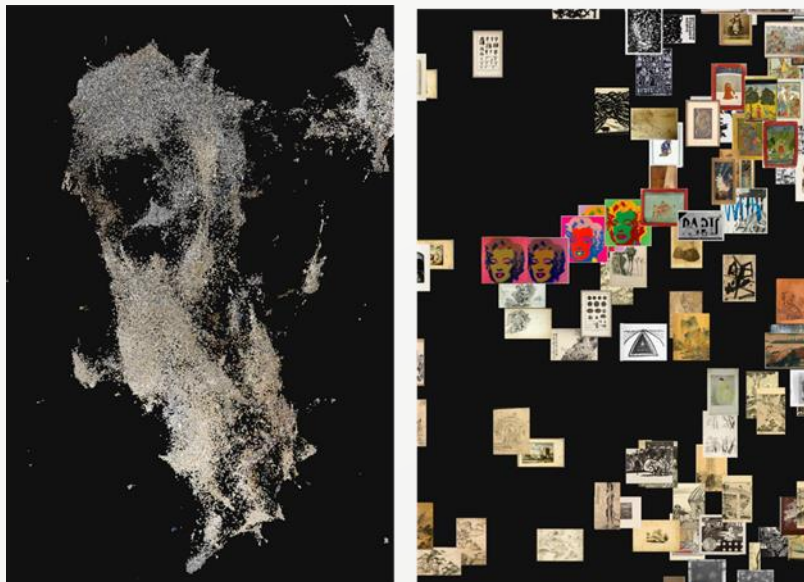


Perceptual empathy

Grounded in **reciprocity** and **familiarity**, perceptual empathy links the dynamics of artistic creation to the digital practices adopted in museums. Over the last decade museums and curators have become fully aware of this potential.

SPATIAL VISUALISATION

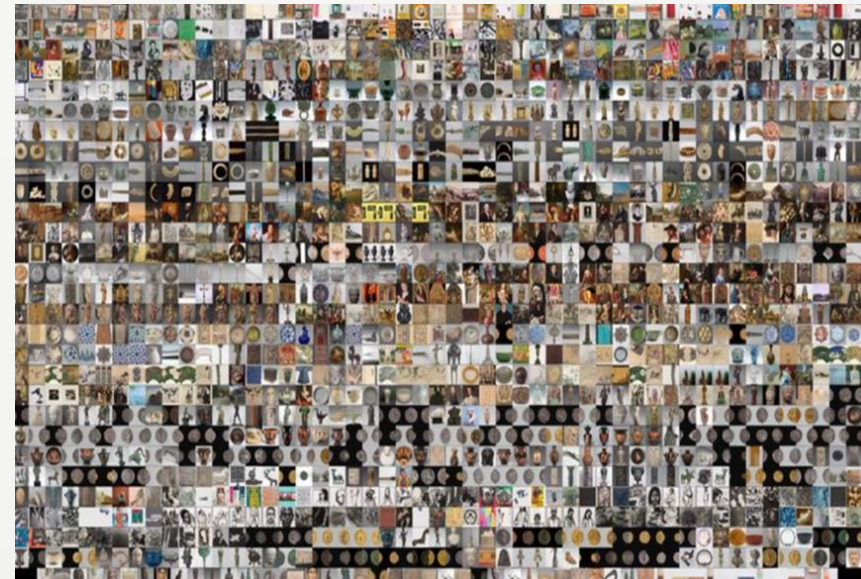
Recognising stylistic motifs to survey an entire collection at a glance.



Dario Rodighiero, Doug Duhaime, Christopher Pietsch, and Jeffrey Schnapp, **Surprise Machines**

GENERATIVE NETWORKS

GANs and neural-data applications shape reflection on historic-artistic collections and their access.

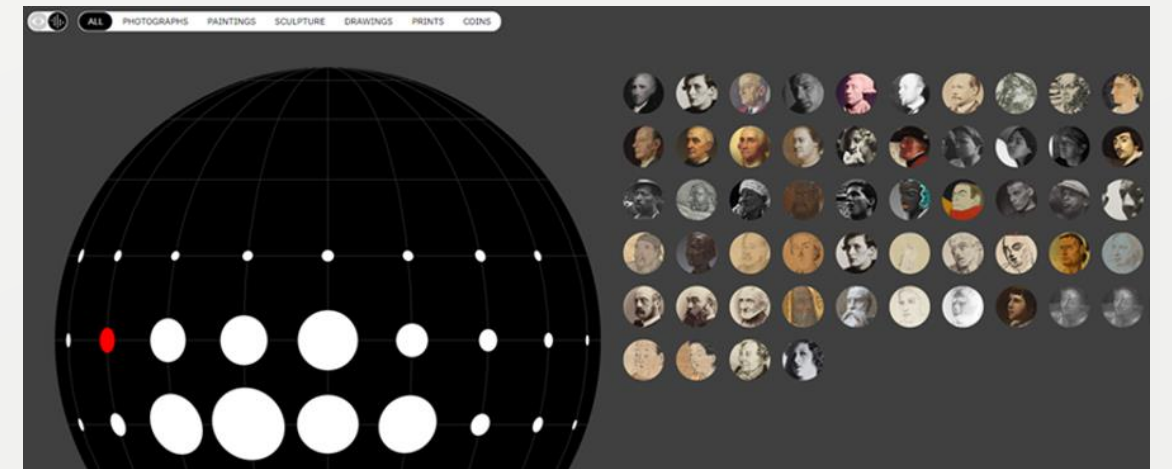


Sinan Goknur/metaLAB, **Object Map**

<https://mlml.io/p/ham-object-map/>

A GLANCE AT THE CATALOGUE

Tools for curator and historian to reason about how works are displayed and used.



Jeffrey Schnapp, Dietmar Offenhuber, Todd Linkner, and Kevin Brewster, **A Flitting Atlas of the Human Gaze**

<https://metalabharvard.github.io/ars-flittingatlasofthegaze/>

Can AI foster awareness of the artwork?

Can artificial intelligence in museums contribute to *awareness* of the artwork — acting as a narrative device of an **ekphrastic** nature? By **awareness** we mean more than information and other than entertainment: the capacity to stand before the work knowing *what* we are looking at, *why*, and through which *mediations*.

AWARENESS

Not information, not entertainment — knowing what we look at, and through which mediations.

ACCESSIBILITY

A narrative that does not lower the threshold of entry, does not widen awareness >Is it true?

AGENCY

Images are never passive recipients of our descriptions.

Mapping AI in museums

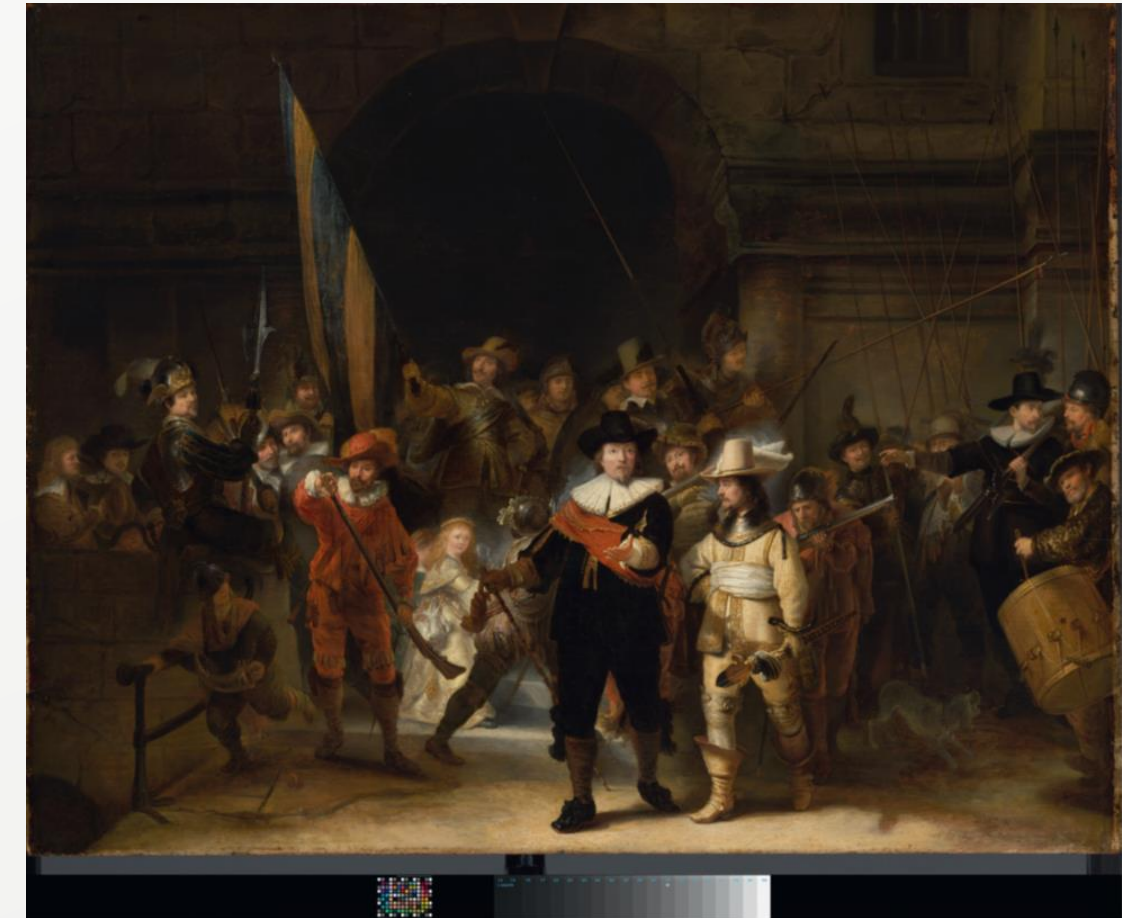
A recent mapping shows that in central Europe the use of AI is directed at the **documentation, research and conservation** of collections — including projects that use neural networks and machine learning to reconstruct works partially or totally destroyed by war or exhibition needs.

- A. Documentation of the collection
- B. Research & interpretation
- C. Conservation & reconstruction

Operation Night Watch

Rijksmuseum, Amsterdam · 2021

An AI model reconstructed the destroyed edges of Rembrandt's *The Night Watch* (1642), trained on datasets of the artist's stylistic features and taking as real reference a 17th-century copy attributed to Gerrit Lundens.



Klimt vs Klimt: the Faculty Paintings

Belvedere × Google Arts &
Culture · 2021

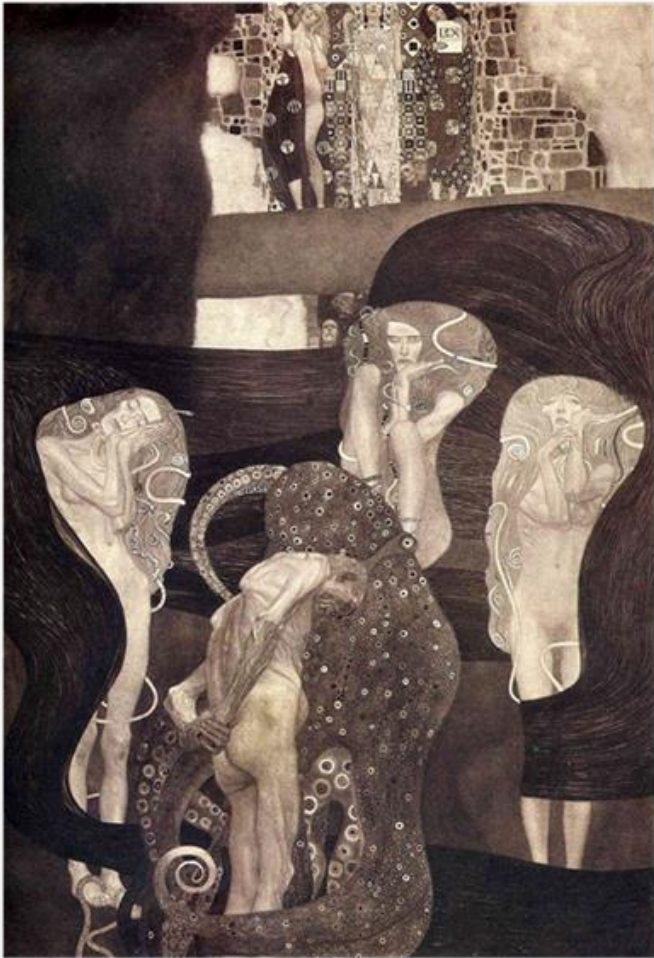
Machine-learning
reconstructed Klimt's three
monumental *Faculty
Paintings* — made for the
1900 Vienna Secession,
destroyed by fire in 1944,
and surviving only as
black-and-white
photographs.



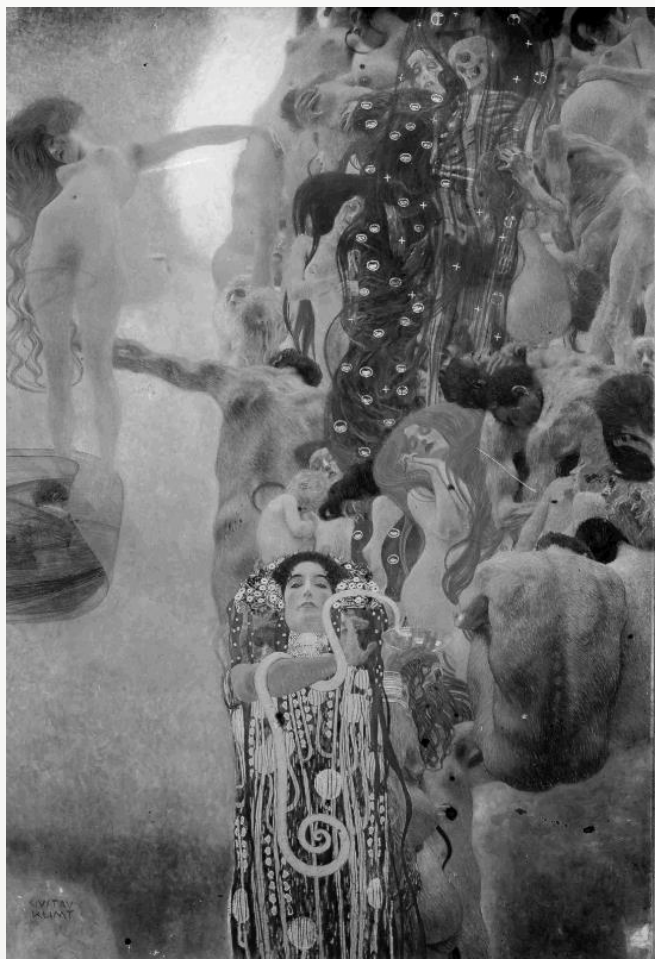
Philosophy, 1899–1907. Ceiling
panel for the Great Hall of Vienna
University



Medicine, 1899–1907



Jurisprudence, 1899–1907



The museum leaves behind its status as a neutral container

- In **AI-mediated** research the museum takes the form of an ***assemblage of objects***, individuals, forms and narratives — echoing Deleuze & Guattari, and above all **Latour's actor-network theory** .
- No longer a **neutral container**, it becomes an ***extension of analytical sensibility*** : a dynamic environment of interconnections between human and non-human actors, restoring the complexity of the material, immaterial and technological presences that inhabit it.
- What finds impulse is the act of **co-creation**.

THE "AI-MUSEUM" — A SOCIO-TECHNICAL ASSEMBLAGE OF

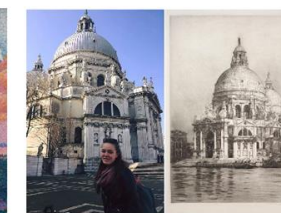
- AI technologies (computer vision, NLP, neural networks)
- Human beings (staff, researchers, IT experts, visitors, artists)
- Material objects (historic artefacts, artworks)
- Real or virtual environments (exhibition spaces, digital archives)

AI Collection Toolsets



AI: Extend the Art

Combine your imagination with the potential of AI to create a world beyond the frame! Developed for educational purposes, this tool converts ideas into artwork, to hone prompt engineering skills while...



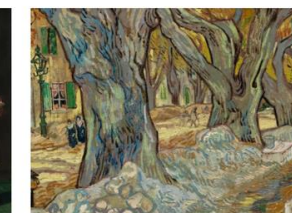
AI: Share Your View

Upload an image from your device to see matches from the collection. If you find a good match, add it to the gallery to share it with the world.



AI: ArtLens for Slack

ArtLens for Slack shares the CMA's world-class collection and lets you and your coworkers curate a new exhibition, unique to your company every work day.



CMA One Minute Guides

Google Arts and Culture worked with the CMA to create a special edition of their One Minute Guides using Google's multimodal AI model Gemini and the CMA's Open Access API. Explore personalized guides...



Digital co-creativity in digital museums

A user-centred approach that engages audiences, both virtual and physical, in **accessing, performing and knowing** a collection. In relation to the art museum, co-creation pursues four purposes:

01

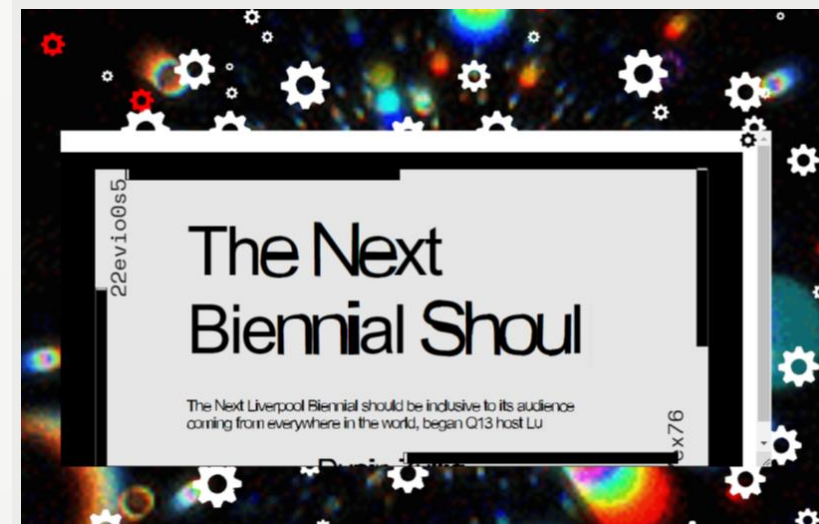
Carnegie Museum of Art



Education

02

Impett & Krysa · UBERMORGEN ·
2021-22
The Next Biennial Should be Curated by a Machine,
2021-12 - <https://biennial.ai/>



Curatorial

03

Curatorial
A(i)gents

Harvard Art Museums · metaLAB
· 2021

CURATORIAL
A(i)GENTS

3.01.22 — 5.15.22

HARVARD ART MUSEUMS, LIGHTBOX GALLERY

Display

04

Lucy Seal, Aidan Meller, Ai-DA, 2019,
Oxford University



Agency

The overall aim is to improve the experience of the museum as a whole.

What images do to us

Before asking what the algorithm does to images, recall **what images do to us**.

For Alfred **Gell**, the artwork is an *index* — a material thing from which the viewer infers, by **abduction**, an *agency* behind and within it. Art is explained not by meaning or pleasure but by a *nexus* binding artist, index, prototype and recipient.

David **Freedberg**: the power of images — efficacy precedes



THE COMPUTATIONAL AGENT

Before a generative model's output we perform the same abduction of agency once reserved for an idol or a portrait.

Now the *index has been computed*, the agency is distributed: prompt-writer, trainers, the millions of absorbed images, the filters that decide the sayable.

Generated images do not arrive stripped of power — they inherit it.

“What would a machine dream about after seeing the collection of The Museum of Modern Art?”

Refik Anadol, Unsupervised, Museum of Modern Art, 2022

From the catalogue to latent space

THE CATALOGUE

Describes by fields — author, date, technique, subject, provenance.

Irises

1889

Vincent van Gogh (Dutch, 1853 - 1890)

On view at [Getty Center, Museum West Pavilion, Gallery W204](#)

[View full record details](#)



In May 1889, after episodes of self-mutilation and hospitalization, Vincent van Gogh chose to enter an asylum in Saint-Rémy, France. There, in the last year before his death, he created almost 130 paintings. Within the first week, he began *Irises*, working from nature in the asylum's garden. The cropped composition, divided into broad areas of vivid color with monumental irises overflowing its borders, was probably influenced by the decorative patterning of Japanese woodblock prints.

There are no known drawings for this painting; Van Gogh himself considered it a study. His brother Theo quickly recognized its quality and submitted it to the *Salon des Indépendants* in September 1889, writing Vincent of the exhibition: "[It] strikes the eye from afar. It is a beautiful study full of air and life."

Each one of Van Gogh's irises is unique. He carefully studied their movements and shapes to create a variety of curved silhouettes bounded by wavy, twisting, and curling lines. The painting's first owner, French art critic Octave Mirbeau, one of Van Gogh's earliest supporters, wrote: "How well he has understood the exquisite nature of flowers!"

+

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⌕

Download image (3000 x 2368 px)

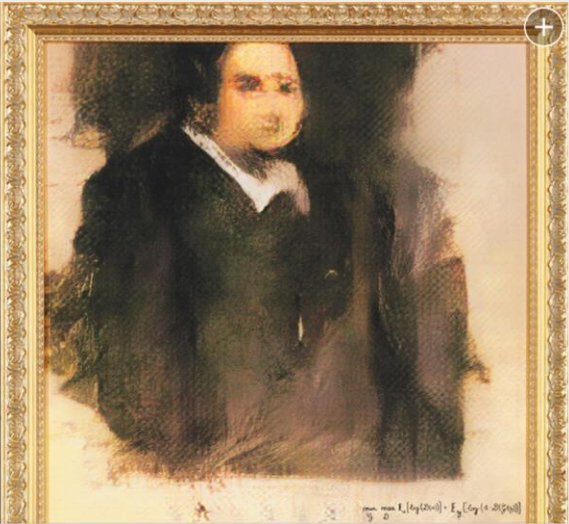
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LATENT SPACE

A mathematical domain that has absorbed a whole visual-cultural repertoire; it **makes non-obvious** relations visible and synthesises the most **plausible** figurations — with them, *hallucinations* : coherent, non-existent detail.



Portrait of Edmond Belamy, 2018, created by GAN (Generative Adversarial Network). Sold for \$432,500 on 25 October 2018 at Christie's in New York. Image © Obvious

Generates.

Two regimes that do not overlap — the first guarantees, the second generates. A **second describer** of the collections has settled in beside the catalogue, trained on them and on much else, that no institution fully controls.



Prompt writing as a reversed ekphrasis

Ekphrasis — the verbal description of an image — returns to the centre of the debate. Writing a prompt is its *reverse* :
a writing that summons an **absent** image.

EKPHRASIS ↔ HYPOTYPOSIS

Manovich & Arielli (2024):
“Hypotyposis means to
outline or to *draft*, which
stresses the aspect of
“generation,” as trying to
induce with words an image
— the forensic artist drawing
from a witness's words.

THE PROMPT

Verdicchio (2024): "a textual
means to stimulate an agent
to create" — poet and
prompter aligned across the
same four functional
elements.

TRANSDUCTION, NOT TRANSLATION

Rodríguez-Ortega (2024), after
Simondon: Transduction: “dynamic
operation by which a potential field of
energy is actualized, moving from one
state to the next in a process that
individuates new materialities.”
the image translates nothing — so it
makes no sense to judge it by *fidelity*.
The gap doesn't close > it is
transduced.

The museum as the new centre of remediation

New media reshape and absorb the old (Bolter & Grusin 1999) .

Text-to-image models push this to an unprecedented scale:

- every generated image is, a **statistical remediation** of millions of prior images — many from museums' own digitisation campaigns.
- The power is there, and grows.
- Often what is missing is the *place that makes it legible*.

Not a barrier — a laboratory.

If algorithmic remediation happens anyway, the museum can become the centre where it is made **conscious** : the generated image traced back to the works it descends from, discussed in its limits, used as narrative instrument — not surrogate.



It uses AI to transform verbally expressed dreams, wishes and fantasies of exhibition visitors into images. The installation echoes Marcel Duchamp's well-known ready-made Fountain (1917).

ZKM, *intelligent.museum*

Wishing Well
Yannick Hofmann
Interactive Installation
2022–2023



From the connoisseur to the remediator

CONNOISSEUR · LONGHI

The educated eye — without it no narrative holds. The foundation.

CONNECTOR · BERGAMO & LOLLINI

Finds and draws the hidden connections across disciplines — from texts to visual translation.

REMIANIATOR

Mediator and curator together — the go-between of catalogue and latent space.

- The remediator does not produce images — they **lead them back**. Does not censor error — **interrogates** it.
- Does not promise transparency — **organises contexts** where opacity becomes discussable.
- The connoisseur of the generative age.



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